Quantum Cryptography Analogy

Aim : To learn how information can be encrypted and sent using the polarization of light.

Apparatus: Laser diode (635nm), Laser electronics for switching between continuous and pulse mode(millisecond time scale), Sensor electronics, power supply units, Polarizing beam-splitter, Half waveplate polarization rotator, Optical breadboard, Opto-mechanical equipment.

Theory: Cryptography, the encryption of messages and data, has always been a fundamental topic in the field of communication. It describes the encryption of data that is rendering a message unrecognizable, ideally making it readable only for the sender and the recipient. This means that the encrypted message is only meaningful if the key to decode it is known. The security of the key is based on either complex underlying algorithms or on practical constraints such as the factorization of large numbers. A wide variety of different methods were developed over the centuries in order to prevent decryption by third parties. All classical cryptography methods have the disadvantage that one can never be sure that the key used for encryption will not be cracked. This fundamental problem however can be solved with the use of quantum physics. Encryption methods using quantum physics have been proposed which can guarantee safety from interception. The experiment here analogus to quantum encryption discusses the BB84 protocol which combines the one-time pad encryption method with the quantum key distribution. The one-time pad method uses a random binary

sequence of 0s and 1s that constitutes a perfect data transmission key. Adding this key to the intended binary message makes the encrypted message a random sequence of 0s and 1s as well. Using the key to decode the encrypted message returns the original message prior to decryption.

Quantum key distribution (QKD) is a cryptographic technique that allows two remote users to establish a secure key (random bit string) between them which can be used further for secure communication, using the principles of quantum mechanics. In 1984, Charles Bennett and Giles Brassard put forward the first quantum cryptography protocol named as BB84 protocol which uses polarization states of photons as the physical entity to generate a secure bit string among two users. A pair of orthonormal polarization bases are required because according to Heisenberg's principle, if the polarization state is defined in one of the bases, it is undefined in the other. The protocol uses such four polarization states. The most important advantage of QKD is that, in principle, the users can detect the presence of an eavesdropper who is trying to gain information from the transmission of information occurring between the sender and the receiver. Since an exact copying of a quantum physics state is actually impossible without altering its state, this principle is generally known as the "no-cloning theorem" which was formulated and proven in 1982. This quantum physics aspect relies on using a single photon light source to carry the information, such that a single information bit is carried by only one photon in a specific state and therefore cannot be copied. However in this experiment, all the results can be described through classical physics with the use of diode lasers making it to work in pulse mode with the help of switching electronics, hence the functioning of this setup is completely analogous to the quantum setup.

The BB84 protocol functions by defining two bases that each include two polarizations of light: the + basis consists of 0^0 and 90^0 polarizations and the x basis consists of -45^0 and 45^0 polarizations. In this scheme, either basis can be

used to represent a binary 0 (0^0 or -45^0) and a binary 1 (90^0 or 45^0). The first person Alice sends a random bit in a random basis and the second person Bob measures it in a random basis. They then exchange the basis via a public channel. If they each used a different basis, the measurement is discarded; if the basis is same, both have now generated a key bit. Since the public exchange only contains the basis, the bit is unknown to others. If the third person Eve attempts to intervene between Alice and Bob, she too can only guess the basis for each bit. Because the basis guess is random, the wrong basis will be chosen in 50% of all cases, which automatically results in errors that Alice and Bob can detect by exchanging few bits.

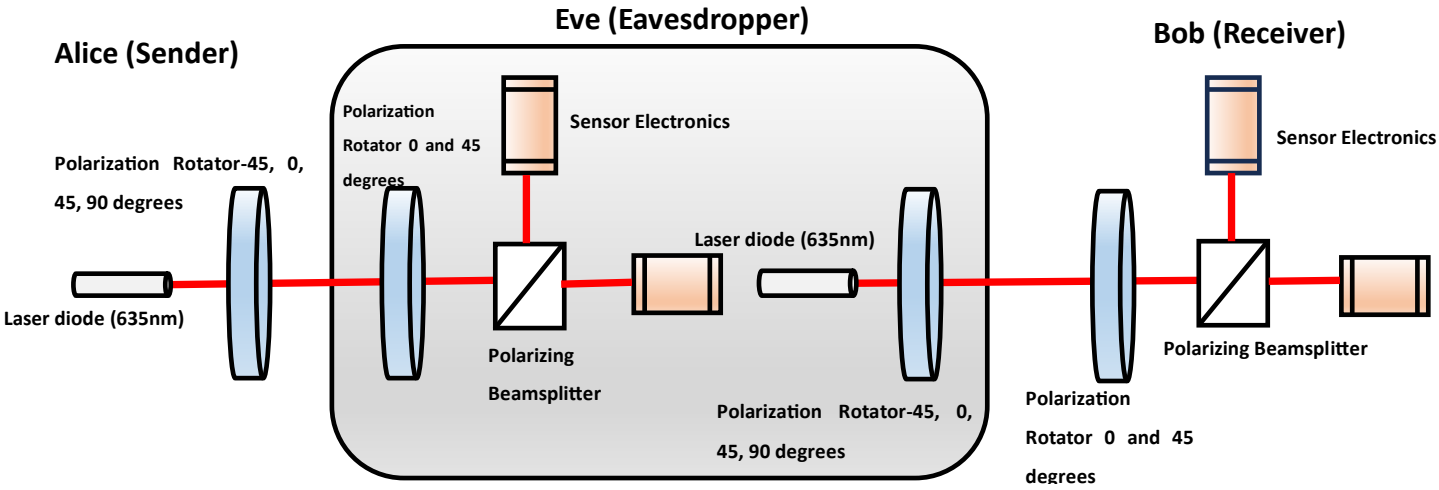

Figure 1 Schematic diagram of Quantum Cryptography Experimental setup.

Experiment Procedure: The experiment is divided into three sections:

1. Generating a key with a length of at least 20 bits.
2. Encryption and transmission of a 4-letter word.
3. Installation of Eve and detection of an eavesdropper.

Key Generation:

- Set up Alice and Bob so that they face each other at a large distance of approximately 60cm.
- Alice, Bob and Eve should have a program in any software to generate random numbers between 0 and 1 and assign the value 0 to generated random numbers which is less than 0.5 and 1 to numbers between 0.5 and 1.
- Alice will generate two sets of such random 0s and 1s of a total of 52 in number.
- Bob will generate one set of such random 0s and 1s of a total of 52 in number.
- The polarization angles -45, 45,0 and 90 degrees are divided into set of two basis and there is a bit associated with each angle in each basis as shown in the table below:

Bits associated	Basis (+)	Basis(X)
0	0^0	-45^0
1	90^0	45^0

- The random 0s and 1s generated by the Alice and Bob will be used in the selection of basis and angles to be set for the transmission of bit corresponding to it.
- The table below shows such selection that should be made.

Random 0s and 1s	Basis selected	Bit selection for transmission
0	Basis(X)	0
1	Basis (+)	1

- Now, Alice will send those 52 bits in her randomly selected basis and angles and Bob will receive the transmitted bits by setting his basis randomly.
- Once, the transmission is completed now Alice and Bob will publically announce the bases for each measurement. They will not share the bits transmitted and received. After that Bob will see which of the bases are matching. They then erase any measurements where the bases do not match.

Encryption and Transmission of an Alphabet:

- The bits of the bases that were matching now forms the encryption key. It should not be share to anyone.
- Alice will now encrypt the message of alphabet chosen with the help of generated key. Alice will get the binary representation of that alphabet by converting its decimal number (place value in terms of serial no. of alphabet eg. A-1, B-2, C-3 etc.) to binary format.
- Alice will now do the binary addition operation on binary representation of alphabet with the generated key. The obtained result will now form the encrypted message in terms of the binary form containing bits 0s and 1s.
- Alice will transmit these encrypted message bits to Bob, by selecting a single basis either + or X and by selecting suitable angle corresponding to bits that needs to be transmitted.
- Before, doing previous step, Alice will inform Bob about the selection of bases that is being used in the transmission of encrypted message. Hence Bob will set the angle corresponding to that basis. Bob will receive the encrypted bits.

- Bob will now decrypt the message send by the Alice. Bob will perform binary addition operation on the received encrypted message bits with key generated earlier. The result of binary addition gives Bob decrypted bits.
- Bob will convert the decrypted bits to alphabet according to binary to decimal conversion.

Installation of Eve and detection of an eavesdropper:

- Place Eve between Alice and Bob, and set both sensor electronics to adjustment mode (LED lights up Yellow). Adjust Eve's receiver so that all 8 transmission cases (in table below) work with Alice. Then, adjust Eve's transmitter and Bob's receiver so that all 8 transmission cases work with Bob. Return to the measurement mode for both sensor electronics (LED lights up green).

Alice			Bob				Same Basis
Basis	Bit	Angle	Basis	Angle	Detector"0"	Detector"1"	check
+	0	0	+	0	100%	0%	Yes
+	1	90	+	0	0%	100%	Yes
X	1	45	+	0	50%	50%	No
X	0	-45	+	0	50%	50%	No
+	0	0	X	45	50%	50%	No
+	1	90	X	45	50%	50%	No
X	1	45	X	45	0%	100%	Yes
X	0	-45	X	45	100%	0%	Yes

- Once, the checking is done then it ensures that everything is working fine.
- Alice, Bob and Eve will generate random nos. between o and 1 and do the same thing as Alice and Bob did earlier. Eve will generate one set of random nos. of total 52 in number.
- These random numbers are being used in the selection of bases and angles.
- Alice will send the bits with the help of those randomly selected bases and angles. Eve will now receive the data by randomly choosing the basis. The bits that Eve will receive in the chosen basis angles will now be used in the transmission to Bob.
- Eve will make sure the settings of polarizer to be such that whatever the bit Eve received should be transmitted to Bob with same bases. It is like getting the information in between by random intruder, copying the data and sending/forwarding the same thing to person ahead without knowing the third person of his/her presence.
- Bob will receive the bits in his randomly selected basis.
- Alice and Bob will now share the basis they used to transmit the data without sharing the information of bit shared.
- They will rule out the bits shared for which the bases is not matching.
- Alice will now share the bits of matched basis to Bob.
- Bob will figure out how many of the data bits are there for which basis is matching but the bits are not matching called as error bit. Bob will figure out the percentage of error bits in the total no of bits for which the bases are matching.
- If it is close to 25% then it indicates that there is some intruder in between the communication. And if it is less than 15% then the data transmission with key generated can be thought of to be secure to transmit.
- How, the observation table will look like is shown below with the help of one example:

Measuring protocol for key generation-Alice

	1	2	3	4	5	6	7	8	9
Basis (+ or X)									
Bit (0 or 1)									

Measuring protocol for key generation-Bob

	1	2	3	4	5	6	7	8	9
Basis (+ or X)									
Bit (0 or 1)									

Generated key - _ _ _ _ _ _ _ _

Table for encryption of the message-Alice

letter																				
Data bit																				
Key bit																				
Encrypted bit																				

Table for encryption of the message-Bob

Received Bit																				
Key Bit																				
Data bit																				
Letter																				

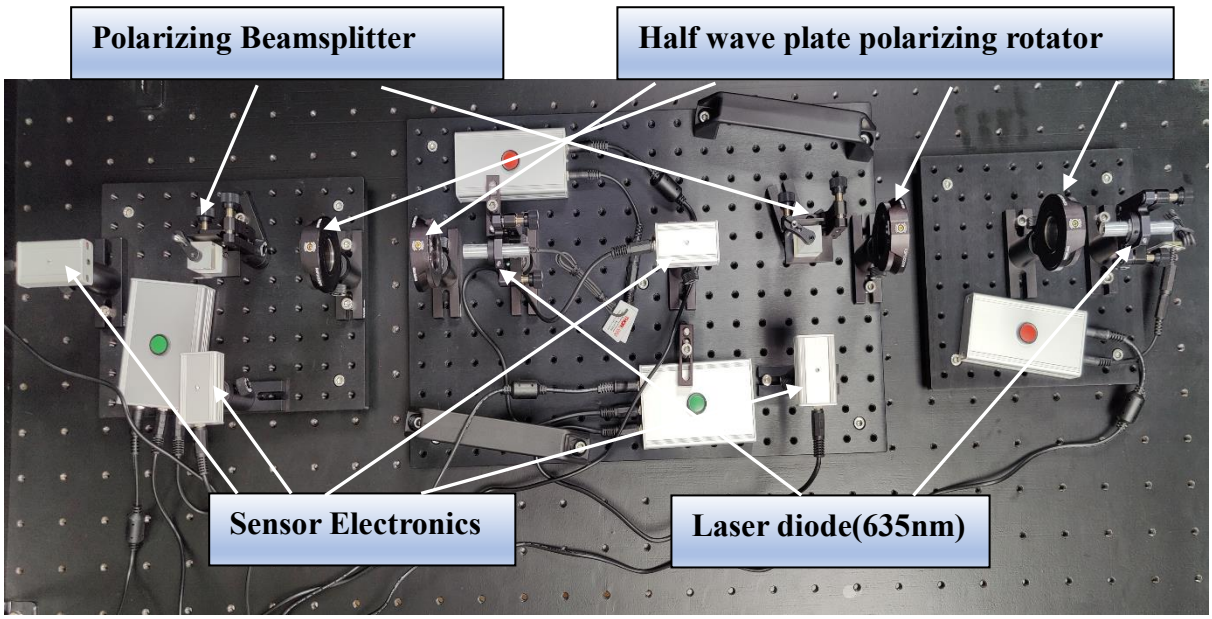

Figure 2 Experimental setup.

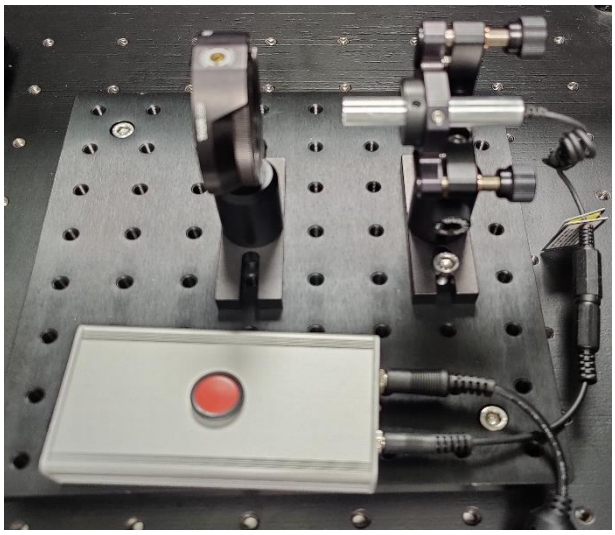

Figure 3 Alice

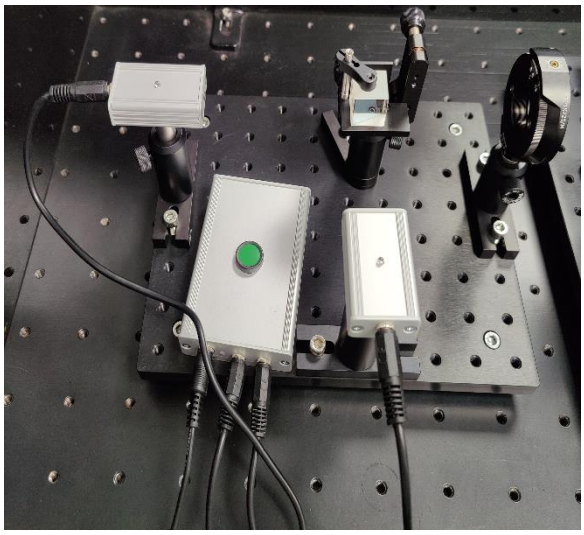

Figure 4 Bob

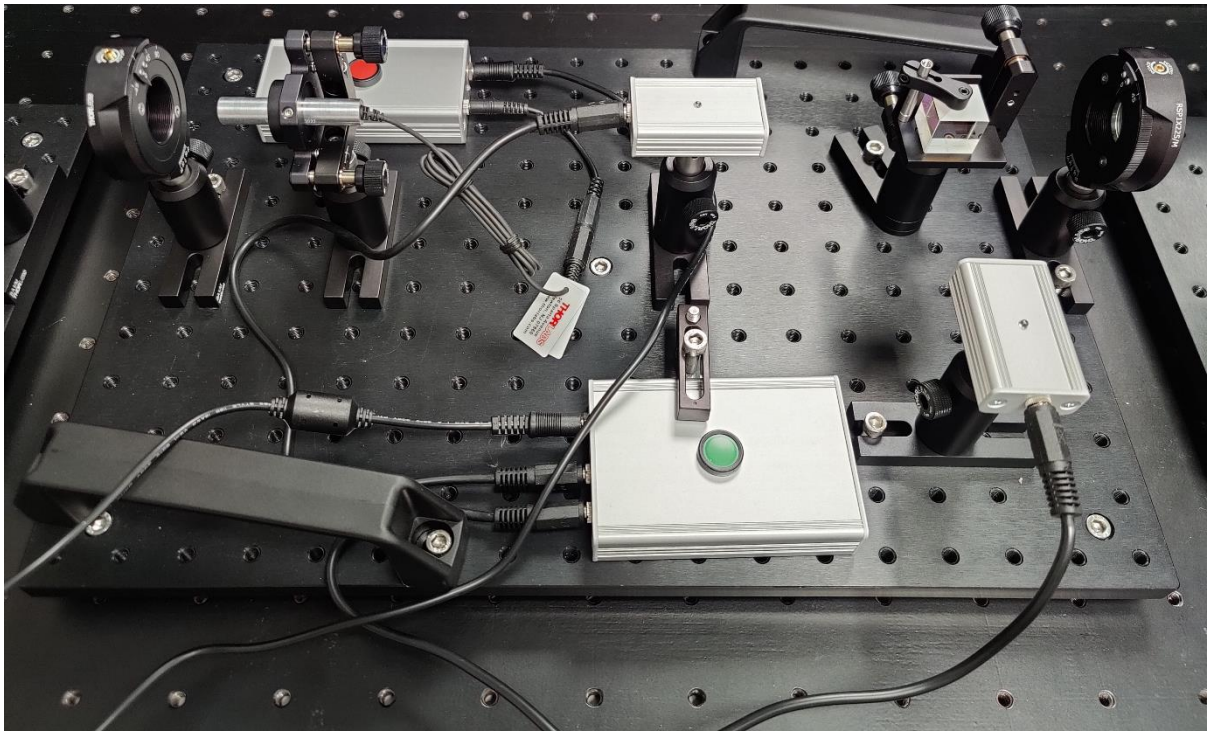

Figure 5 Eavesdropper

Precautions:

- Always make sure that the laser should not fall directly into your eyes. Always maintain some distance from the laser.
- Do not touch the polarizing beam splitter with bare hand.
- Rotation and adjustment of polarization marking values on half wave plate should be done slowly.
- Always make sure the power supply cable is connected properly to socket and the electronics properly.
- Movement of the bread board on which Eve is installed should be done carefully.
- Once the experiment is completed cover all the optical equipments with the plastic zip lock.

Questions:

- Write in short about the working principle of the pulsed lasers.
- What is the material of the polarizing beam splitter. Explain its working principle.
- Show that the polarization vectors in one particular basis can be written as a linear combination of vectors in other basis. What is the probability that a 0 deg polarized photon passes a 45 deg oriented polarizer.
- In terms of bits what percentage of the total data points, the bits of Alice and Bob will be matching, considering Eve is not present.
- If the polarization of light falling on the half wave plate makes an angle of 22.5 deg with the optic axis of the half wave plate, this light after passing through the half wave plate falls on the polarizing beam splitter. What would be the difference between the transmitted and reflected light in terms of intensity of light after passing through polarizing beam splitter.